SESSION 4

# Principles of Responsible AI

*Six Principles · Real-World Evidence · GeoAI Tie-Ins · The Global & Malaysian Regulatory Picture · Ethics Card Sorting*

Day 1 · 7 July 2026 · GeoAI Hackathon Programme

***How to use these notes.*** *These notes are self-contained: every principle is explained with its landmark real-world case, a GeoAI tie-in for your project, plus added context on how these principles now appear in international and Malaysian AI governance frameworks.*

## Learning Objectives

By the end of this session, you should be able to:

- State and explain the six principles of Responsible AI.
- Support each principle with a documented real-world case.
- Translate each principle into a concrete GeoAI design decision for your hackathon project.
- Describe how the same principles appear in the OECD AI Principles, the EU AI Act, and Malaysia's National Guidelines on AI Governance & Ethics (AIGE).
- Classify ethical scenarios under the correct principle (Activity 4).

## The Six Principles at a Glance




*Figure 4.1 — The six principles of Responsible AI. Real systems must satisfy all six at once.*

Each principle below follows the same three-part structure: what it means, the landmark real-world case that proves it matters, and the GeoAI tie-in — the concrete design decision it implies for your hackathon project.

### Principle 1 · Fairness

**Meaning.** AI outcomes should not systematically help or harm one group more than another. The system should work about as well for everyone it touches — not just the group best represented in its training data.

**Landmark case:** a 2019 study published in Science examined a US healthcare algorithm used on roughly 200 million patients a year. It used healthcare cost as a stand-in ("proxy") for illness. Because less money was historically spent treating Black patients with the same conditions, the algorithm concluded they were healthier than they really were — cutting the number flagged for extra care by more than half. The deeper lesson: bias entered through an innocent-looking proxy variable, not through any explicit reference to race (Obermeyer et al., Science, 2019).

**GeoAI tie-in:** a crop-health model trained only on commercial-farm imagery can look "accurate" overall while quietly failing every smallholder farm. Check accuracy per group, not just in total.

## Principle 2 · Privacy

**Meaning.** Personal and location data should be collected, stored, and shared only with people's knowledge and consent. "Anonymised" data is not automatically safe data.

**Landmark case:** in 2018, fitness app Strava published a global heatmap built from an "aggregated and anonymised" view of over a billion activities. It still clearly outlined the perimeters and patrol routes of secret military bases in Iraq, Syria, and Afghanistan — soldiers were often the only people jogging in those exact spots. Aggregation failed precisely where data was sparse.

**GeoAI tie-in:** the same logic applies to any GeoAI dashboard built on sparse rural data — a handful of points can expose exactly who lives where. Sparse-area data needs stronger masking, not weaker.

## Principle 3 · Transparency

**Meaning.** People affected by an AI decision should be able to understand, in plain terms, roughly how and why it was reached — not just told to trust a score.

**Landmark case:** in State v. Loomis (2016), a Wisconsin court sentenced a defendant partly on a proprietary COMPAS recidivism-risk score. Because the algorithm was a trade secret, neither the defendant nor the court could examine the calculation behind it — yet the court upheld its use anyway. The case remains the standard citation for why unexplainable scores and high-stakes decisions are a dangerous mix.

**GeoAI tie-in:** if a flood-risk model can't explain which input — rainfall, elevation, past reports — drove a "high risk" label, no one can catch it when it's wrong. Techniques like SHAP (see Session 1's KL heat-island case) exist precisely for this.

## Principle 4 · Accountability

**Meaning.** Someone specific must be responsible for an AI system's decisions and answerable when it causes harm — "the algorithm did it" cannot be the end of the conversation.

**Landmark case:** in 2018, an Uber self-driving test car struck and killed pedestrian Elaine Herzberg in Tempe, Arizona — the first pedestrian death involving an autonomous vehicle. The NTSB investigation found the system detected her 6 seconds before impact but couldn't classify her; Uber had disabled the car's automatic emergency braking; and the human backup driver was distracted. No single point of ownership existed — which is exactly what enabled the failure.

**GeoAI tie-in:** before a disaster model goes live in your hackathon demo, name — out loud — who signs off on it and who is accountable if it is wrong.

## Principle 5 · Safety & Reliability

**Meaning.** A system should perform consistently under real-world conditions, and fail loudly and safely rather than confidently giving a wrong answer.

**Landmark case:** Zillow's home-pricing algorithm kept assuming prices would keep rising through 2021 even as the market cooled — a failure of reliability under changing conditions (known as "distribution shift"). The company bought thousands of homes at inflated, algorithm-driven prices, then wrote down over $500 million and cut 2,000 jobs when it shut the programme down.

**GeoAI tie-in:** a flood model that is silently wrong 20% of the time is more dangerous than one that visibly flags "low confidence" — design your system to be able to say "I'm not sure."

### Principle 6 · Sustainability

**Meaning.** Weigh the real environmental and social cost of building and running a model against the value it actually delivers — bigger is not automatically better.

**Landmark case:** a 2019 University of Massachusetts Amherst study found that training one large natural-language AI model (with neural-architecture search) can emit over 626,000 lbs of $CO_2$ — roughly five times the lifetime emissions of an average car, including its manufacture (Strubell et al., ACL 2019).

**GeoAI tie-in:** for most hackathon problems, a smaller, well-tuned model on good local data beats a giant general-purpose one — cheaper to run and kinder to the planet. Using a pretrained model (Session 1) rather than training from scratch is itself a sustainability decision.

## Quick-Reference Summary

| Principle | One-line test for your project | Landmark case |
|---|---|---|
| Fairness | Does it work about as well for every group it touches? | US healthcare algorithm (2019) |
| Privacy | Is data collected with knowledge and consent — and truly non-identifying? | Strava heatmap (2018) |
| Transparency | Can an affected person understand roughly why a decision was made? | State v. Loomis / COMPAS (2016) |
| Accountability | Is one named person answerable if it causes harm? | Uber self-driving fatality (2018) |
| Safety & Reliability | Does it fail loudly and safely rather than confidently wrong? | Zillow Offers write-down (2021) |
| Sustainability | Is the environmental cost justified by the value delivered? | UMass Amherst $CO_2$ study (2019) |

## The Bigger Picture — These Principles Are Becoming Law and Policy

The six principles are not just this programme's opinion. They form the backbone of the world's major AI governance frameworks — which means practising them in your hackathon is practising for the real regulatory environment your project would face.

### International Frameworks

- **OECD AI Principles:** the first intergovernmental AI standard, calling for AI that is values-aligned, transparent, robust, and accountable, with capacity for human agency and oversight, plus systematic risk management across the AI lifecycle. The OECD framework directly influenced the EU AI Act, the US AI Bill of Rights, and the UK's AI White Paper.

- **EU AI Act:** the world's first comprehensive legal framework for AI, in force since 1 August 2024 and applying in phases (most high-risk obligations from 2026). It classifies systems into four risk tiers, with strict duties — risk management, data quality, documentation, and human oversight — for high-risk systems.

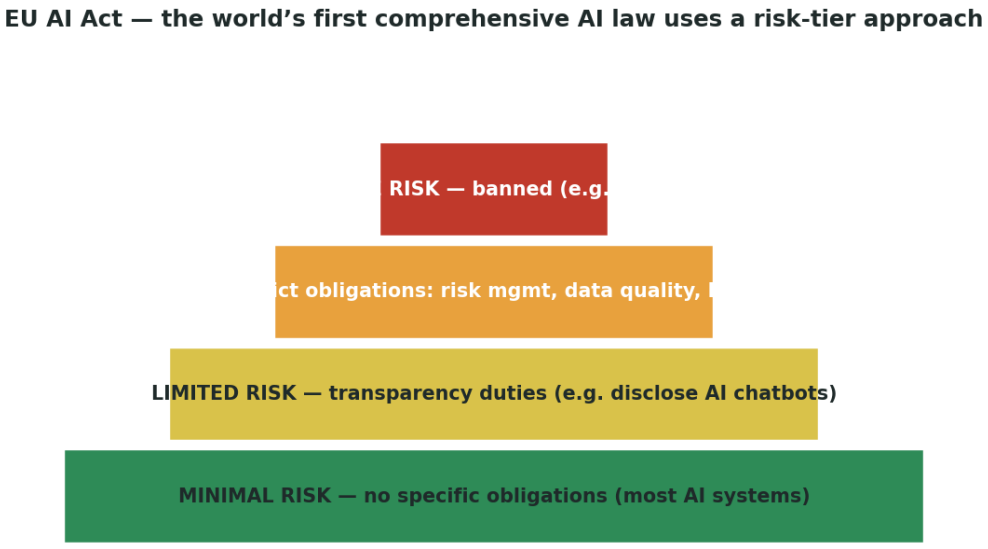



*Figure 4.2 — The EU AI Act's four risk tiers. Obligations scale with risk; most safety-critical GeoAI would be treated as high-risk.*

- **Human oversight, made explicit:** Article 14 of the EU AI Act requires high-risk systems to be designed so humans can effectively oversee them — including the ability to understand the system's limits, to detect anomalies, to guard against automation bias (over-relying on machine output), to correctly interpret outputs, and to intervene or stop the system safely. This is the origin of the seventh "Human Oversight" dimension on your hackathon checklist (Session 6).

## Malaysia's Framework — the AIGE

**Directly relevant to you:** on 20 September 2024, Malaysia's Ministry of Science, Technology and Innovation (MOSTI) launched the National Guidelines on AI Governance & Ethics (AIGE) — among the first national-level AI governance guidelines in ASEAN. The AIGE is voluntary (there is not yet a dedicated AI law in Malaysia) and supports the National AI Roadmap (AI-RMAP 2021–2025), with guidance for three stakeholder groups: end users, policymakers, and developers.

The AIGE sets out seven core principles, adapted from UNESCO and OECD standards — notice how closely they mirror this session:

| AIGE principle (Malaysia, 2024) | Matches this session's principle |
|---|---|
| Fairness | Fairness |
| Reliability, Safety and Control | Safety & Reliability (+ Human Oversight, Session 6) |
| Privacy and Security | Privacy |
| Inclusiveness | Fairness (extended: benefits reach all groups) |
| Transparency | Transparency |
| Accountability | Accountability |
| Pursuit of Human Benefit and Happiness | Sustainability (extended: net value to society) |

Related Malaysian developments worth knowing: the National AI Office (NAIO) provides institutional leadership; the Cybersecurity Act (2024) and the revised Personal Data Protection Act (PDPA) strengthen the surrounding legal framework; and at regional level, ASEAN member states approved the ASEAN Guide on AI Governance and Ethics in February 2024.

Pitch tip: telling judges "our design follows the AIGE's seven principles" — and showing exactly how — signals professional-grade thinking that most hackathon teams never demonstrate.

## Hands-On Activity 4 — Ethics Card Sorting (8 minutes)

**Task:** classify each scenario card under the principle it relates to most. Cases involve flood prediction, healthcare, and smart cities.

| Card | Scenario |
|---|---|
| A | A hospital AI triages patients faster for wealthier postcodes. |
| B | A flood app shares exact home coordinates with a third-party advertiser. |
| C | A smart-city model can't explain why it flagged a neighbourhood as high-risk. |
| D | No one is responsible when a disaster alert fails to send. |

Workspace: write each scenario letter (A–D) under the matching principle — Fairness · Privacy · Transparency · Accountability · Safety · Sustainability.

Suggested mapping to discuss: A → Fairness · B → Privacy · C → Transparency · D → Accountability. Be ready to defend overlaps — many real failures touch several principles at once (D is also a Safety failure; A also raises Transparency questions about the triage logic).

## Glossary

| Term | Meaning |
|---|---|
| Proxy variable | An innocent-looking stand-in (e.g. healthcare cost for illness) through which bias can enter a model. |
| COMPAS | A proprietary US recidivism risk-scoring tool at the centre of the State v. Loomis transparency case. |
| Automation bias | The human tendency to over-trust and over-rely on machine outputs. |
| Distribution shift | When real-world conditions drift away from the data a model was trained on (Zillow's failure mode). |
| OECD AI Principles | The first intergovernmental AI standard; basis for much later regulation. |
| EU AI Act | The world's first comprehensive AI law (in force Aug 2024); four risk tiers with duties scaled to risk. |
| AIGE | Malaysia's National Guidelines on AI Governance & Ethics (MOSTI, Sept 2024) — seven voluntary principles. |
| AI-RMAP | Malaysia's National AI Roadmap 2021–2025, which the AIGE supports. |
| PDPA | Malaysia's Personal Data Protection Act — the general data-protection law relevant to location data. |

## Self-Check Questions

1. In the US healthcare case, no one told the model about race — so how did racial bias get in? Name the mechanism.
2. The Strava data was aggregated and anonymised, yet still leaked military bases. What property of the data made aggregation fail?
3. Why is “fail loudly and safely” better than “fail rarely but silently” for a flood model? Use the Zillow case in your answer.
4. Match each of the AIGE’s seven Malaysian principles to the closest principle in this session.
5. Under EU AI Act thinking, which risk tier would a flood early-warning system most likely fall into, and name two obligations that would follow.

## Key Takeaways

- Six principles define Responsible AI: fairness, privacy, transparency, accountability, safety & reliability, and sustainability.
- Each principle is backed by a documented, high-cost failure — these are engineering constraints, not abstractions.
- Every principle converts into a one-line test you can run against your own hackathon project.
- The same principles now anchor the OECD Principles, the EU AI Act (with human oversight required by law for high-risk systems), and Malaysia’s own AIGE — so practising them is practising for the real world.
- Real incidents usually violate several principles at once; classifying them sharpens your ability to spot risk early.

## Sources & Further Reading

- Obermeyer et al., Science (2019); Washington Post / CNN — Strava heatmap coverage (2018); State v. Loomis, Wisconsin Supreme Court (2016).
- NTSB investigation — Death of Elaine Herzberg (2018); CNN / Stanford GSB — Zillow Offers (2021); Strubell et al., ACL (2019).
- OECD — AI Principles (oecd.org/en/topics/ai-principles); EU AI Act, incl. Article 14 on Human Oversight (artificialintelligenceact.eu).
- MOSTI — National Guidelines on AI Governance & Ethics (AIGE, 2024); UNESCO Global AI Ethics Observatory — Malaysia country page; ASEAN Guide on AI Governance and Ethics (Feb 2024).
- Microsoft Responsible AI Standard; ISO — Building a Responsible AI; Harvard DCE — 5 Key Principles of Responsible AI.